

Tab 1

BUSINESS REQUIREMENT DOCUMENT

DZONGJUK (DSTS)



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DZONGKHA STANDARD TESTING SYSTEM



LANGUAGE RESEARCH AND PROMOTION DIVISION
DEPARTMENT OF CULTURE AND DZONGKHA DEVELOPMENT
MINISTRY OF HOME AFFAIRS

System Name:
DZONGJUK (DSTS)

Ministry/Agency Name:
Ministry of Home Affairs

Department Name:
Department of Culture and Dzongkha Development,

Division Name:
Language Research and Promotion Division

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1. INTRODUCTION

1.1 PURPOSE

The purpose of this Business Requirements Document (BRD) is to define the comprehensive functional and technical requirements for developing an integrated digital system for the Dzongkha Standard Testing System (DSTS)-Dzongjuk. This document serves as the foundational blueprint for designing, developing, and implementing an end-to-end automated platform that will manage all aspects of Dzongkha language assessment-from test registration and administration to evaluation, scoring, and result dissemination.

This BRD is intended to:

- Establish a clear understanding of business needs and system objectives among all stakeholders including Language Research and Promotion Division, implementing institutions, system developers, and end users
- Provide detailed specifications for system features, functionalities, and technical architecture
- Serve as a reference document for project planning, development, testing, and quality assurance activities
- Define the scope boundaries to ensure focused development efforts and effective resource allocation
- Facilitate communication and alignment between business stakeholders and technical implementation teams
- Establish success criteria and performance metrics for system evaluation

1.2 BACKGROUND

Bhutan's cultural identity, sovereignty, and social cohesion are deeply rooted in the Dzongkha language. However, there currently exists no unified national standard for assessing Dzongkha proficiency. Multiple institutions conduct independent examinations, resulting in inconsistent standards, duplicated efforts, and inefficient use of public resources.

To address this, the Department of Culture and Dzongkha Development introduced the DSTS, drawing upon global best practices such as IELTS and CEFR, and local assessment frameworks. The growing national importance of Dzongkha across governance, education, citizenship, media, judiciary, and public service demands a robust digital system capable of supporting standardized, scalable, and sustainable assessment operations.

1.3 OBJECTIVES

The objectives of the system are to:

- Provide a centralized digital mechanism to **conduct DSTS Test** and manage examination results, band scores, and certificates.
- Ensure timely, accurate, and transparent publication of results.
- Enable controlled access to band scores and result status for test takers.
- Digitize the storage and access of issued certificates for long-term reference.
- Improve administrative efficiency and reduce dependency on manual record-keeping.
- Support re-evaluation outcomes and approved score revisions within the system.

The project has been structured into two phases to ensure the timely development, stabilization, and enhancement of the system.

Phase 1 focuses on the development and implementation of the core functionalities of the **DSTS**. The following components are included within the scope of Phase 1:

1. **User Registration and Profile Management**
 - Online registration for general users.
 - User profile creation, management, and updating of personal details.
2. **Examination Result Management**
 - Provision for authorized officials to update and manage band scores.
 - Result publication and dissemination to registered users through the system.
3. **Certification Management**
 - Automated generation of certificates upon successful completion of the assessment.
 - Secure access for users to download certificates.
 - Certificate verification mechanism for authenticity validation.
4. **Question Bank Management**
 - Storage and management of examination papers and question sets.
 - Administrative access for uploading and maintaining question repositories.

Phase 2 will focus on enhancing the system by introducing additional functionalities to improve user experience and support digital examination processes. The following components are proposed for Phase 2:

1. **Mock Test Module**
 - Practice tests for candidates to familiarize themselves with the examination format.
 - Automated scoring and feedback for mock test attempts.
2. **Online Examination Module**
 - Conducting Dzongkha proficiency examinations through an online platform.
 - Secure login and controlled examination environment.
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3. Examination Monitoring and Audit Logs

- Activity tracking during examinations for monitoring purposes.

This phased approach will allow the project team to first stabilize the core system functionalities and ensure successful deployment, while providing scope for future enhancements and expanded digital services in Phase 2.

1.4 SCOPE

1.4.1 IN-SCOPE

The following functionalities are included within the scope of the DSTS – Dzongjuk system:

1. Registration (General User)

- Online registration
- Profile management.

2. Examination Management

- Provision to update and manage band scores.
- Post-result publication and dissemination.

3. Certification

- Automated certificate generation upon successful completion of the assessment.
- Secure access and verification of issued certificates.

4. Question Bank Management

- Storage of exam paper

1.4.2 OUT-SCOPE

The following functionalities are excluded from the scope of this phase:

- Biometric authentication or verification.
- Online examination or remote test delivery (examinations will be conducted offline).

1.5 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

DCDD	Department of Culture and Dzongkha development
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DSTS	Dzongkha Standard Testing System
CEFR	Common European Framework of Reference for Languages

2. STAKEHOLDERS

The following are the stakeholders of the Dzongjuk system:

1. Citizens/General Users: Primary beneficiaries and end-users of the system, using it to access information and services related to DSTS
2. DCDD: The Department of Culture and Dzongkha Development (DCDD), as the implementing agency of the Dzongkha Standard Testing System (DSTS), is responsible for the overall coordination and implementation of the system. DCDD's roles include planning and executing Dzongjuk activities, developing and standardizing question papers, and overseeing system maintenance and updates in collaboration with technical partners.
3. NDI: NDI integration enables secure, reliable, and seamless verification of users' identities across systems, reducing duplication and manual processes. By integrating NDI in a DSTS system can ensure accurate candidate identification, prevent impersonation and misuse, improve data integrity, and streamline registration and authentication processes.
4. GovTech : Plays a vital role in the development of the Dzongkha Standard Testing System (DSTS) by delivering technical leadership and support across the entire system lifecycle. Its responsibilities include the design and development of the DSTS digital platform, ensuring security, scalability, and interoperability, integrating key government digital services such as NDI, providing ongoing technical maintenance and system upgrades and offering technical guidance to DCDD to ensure the system remains reliable, user-friendly, and sustainable.
5. DCRC: The Department of Civil Registration and Census supports the DSTS system by enabling identification and verification through access to individual registration data and by providing the necessary information or data required for user registration and result generation.
6. RIM: Royal Institute of Management acts as an advisory and support agent for the DSTS during its development stage, providing expertise on exam features, conduction protocols, and all rules and regulations related to the examination. They also support maintaining exam integrity and offer recommendations on best practices for administration, security, and candidate management.
7. RCSC: The Royal Civil Service Commission (RCSC) plays a pivotal role in implementing Dzongjuk within the civil service by conducting Dzongkha Competency Training for civil servants in collaboration with the Department of Culture and Dzongkha Development (DCDD). It makes Dzongjuk a mandatory examination for aspiring Administrative Assistants and institutionalizes Dzongjuk by

rolling it out across key civil service functions, including recruitment, selection, and promotion, thereby strengthening professionalism, standardization, and Dzongkha usage within the civil service

8. ECB: The Election Commission of Bhutan (ECB) plays an important role in implementing Dzongjuk by adopting the Dzongkha Standard Testing System (DSTS) in lieu of the Functional Literacy and Possession of Skills Test currently conducted in electoral processes. In collaboration with the Department of Culture and Dzongkha Development (DCDD), the ECB can develop DSTS question sets tailored to its specific requirements, thereby ensuring consistency, clarity, and accuracy in the use of Dzongkha.

3. CURRENT STATE (AS-IS) & ISSUES

Currently, the Dzongkha Standard Test processes are carried out **manually**. Registration, verification, examination coordination, evaluation, band score compilation, and certificate issuance are managed without a centralized digital system. Application data and supporting documents are collected and reviewed manually, making it difficult to control registration timelines, enforce capacity limits, prevent duplication, and maintain accurate records.

All coordination and communication among stakeholders-including DCDD, exam heads, committee members, and test takers-are conducted through email and phone calls. Notifications regarding application status, examination details, results, and appeals are shared manually, leading to delays, inconsistencies, and increased administrative workload. The absence of an integrated system limits traceability, transparency, and operational efficiency across the entire examination lifecycle.

Pain Points

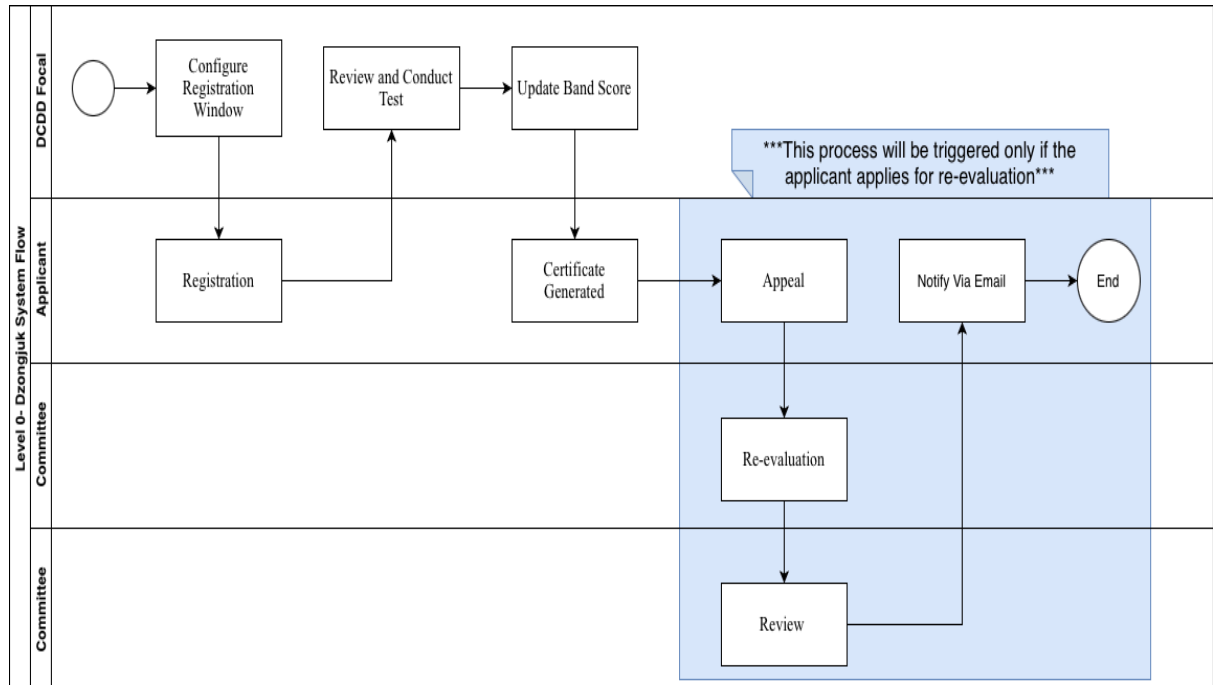
1. High administrative burden due to fully manual processes for registration, verification, evaluation, and certification.
2. Lack of centralized control over registration timelines, capacity limits, and applicant data, leading to inefficiencies and duplication.
3. Delayed communication caused by reliance on email and phone calls for notifications, approvals, and coordination.
4. Limited traceability and auditability of applications, evaluations, approvals, and result declarations.
5. Risk of data inconsistency and human error during manual data entry, score compilation, and record maintenance.
6. Inefficient handling of waiting lists, cancellations, and re-evaluation requests.
7. No secure digital repository for question papers, answer sheets, band scores, and certificates.
8. Poor transparency for test takers, who have limited visibility into application status, results, and certification processes.

4. SYSTEM OVERVIEW (TO-BE STATE)

The Dzongkha Standard Testing System (DSTS) - Dzongjuk will be a digital platform managing all aspects of Dzongkha language assessment. The system will automate candidate registration, application verification, question paper and answer sheet management, evaluation, band score entry, result declaration, and certificate issuance. It will provide a centralized and secure repository for all examination-related data, ensuring consistency, accuracy, and traceability across processes.

The system will eliminate the current reliance on manual procedures, email, and phone communication by introducing role-based access controls and real-time notifications. Test takers will be able to submit applications online within configurable registration windows, track the status of their application, and download certificates securely. DCDD and exam authorities will be able to manage registration timelines, monitor test capacity, verify applicant information, upload question papers and answer sheets, and oversee evaluation and band score submission. A formalized re-evaluation and appeal process will ensure controlled updates to scores, approved only by the Chief of Examiner. By providing automated notifications, audit trails, and accessible reference materials, the system will improve operational efficiency, transparency, accountability, and stakeholder satisfaction while addressing all pain points of the current manual process.

Level 0-Process Flow



Services List

Sl.No	Capability	Services Name	Feature
1	Regulatory and Compliance: Certificate Management	Dzongkha Standard Testing Service	Registration of Applicant
2			Band Score Update
3			Update Attendance
4			Re-evaluation
5			Certificate Generation

1. Role-Based Access Control (RBAC):

- Users are assigned roles such as Author, Reviewer, In-house Editor, Managing Editor, Publisher, Uploader, and System Administrator.
- Access to modules and functionalities is restricted based on user roles.
- One person can have multiple roles.

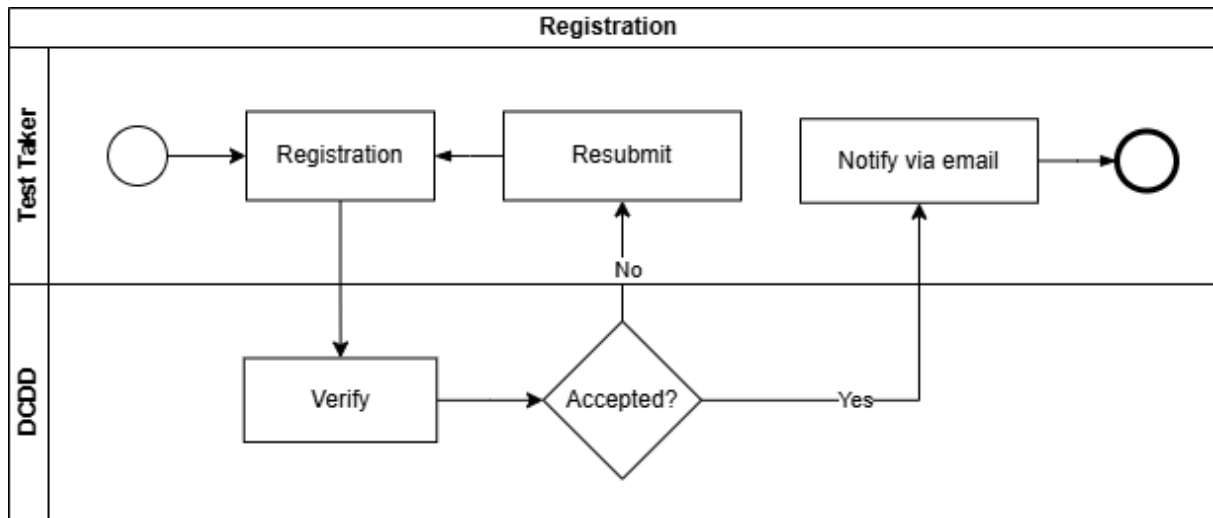
5. BUSINESS REQUIREMENTS

5.1 MASTERS

BR	Requirement
1	Provision to configure and manage the registration timeline and set limits of the test takers.
2	Provision to set the validity for the certificate and be able make changes to the certificates.
3	Provision to set payment amount

5.2 REGISTRATION

5.2.1 PROCESS FLOW



The system shall allow DCDD to configure and control the registration window, including the start and end period and limit for accepting applications. Once the registration window is open, test takers shall be able to submit applications. The system shall capture all required applicant information and supporting documents and ensure all mandatory fields are completed before permitting submission.

Upon successful submission, the system shall automatically forward the application to DCDD for verification and review, and the test taker shall receive an acknowledgement notification confirming receipt of the application. DCDD shall review and verify the submitted application and documents within the system. Based on the outcome of the verification, the application shall be approved or returned to the test taker for correction. After the verification process is completed, the system shall notify the test taker of the final application status with a unique registration number.

5.2.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
BR-1: REGISTRATIONS APPLICATION SUBMISSION (TEST TAKER)	
1	The system shall allow all test takers to submit registration applications only when the registration window is opened and configured by DCDD.
2	Test takers shall be able to enter and submit all required information and supporting documents through the system for registration.
3	The system shall automatically fetch and verify applicant information from external systems, including NDI and census data, to ensure accuracy and authenticity of the submitted information.

4	Test takers shall have the option to cancel their registration at any time before DCDD verification begins. Once the application enters the verification stage by DCDD, cancellation by the test taker shall no longer be permitted. Upon cancellation, the system shall automatically pull up the next applicant from the waiting list, based on the configured maximum capacity limit.
5	Upon successful submission, the system shall provide immediate acknowledgement to the test taker confirming receipt of the application.
6	The system shall automatically check for duplicate registrations during the application submission process. If a test taker attempts to register for the same examination window using the same CID or other unique identifiers, the system shall reject the duplicate application and display an error message informing the test taker that they have already registered for the current examination.

BR NO	BUSINESS LOGIC
BR-2: VERIFICATIONS & APPROVALS (DCDD)	
1	The system shall allow DCDD to configure and manage the registration window, including setting the start and end dates for registration. DCDD shall be able to define the maximum number of registrations allowed within the registration window. Applicants who fall within the allowed registration limit shall be marked in yellow, while those exceeding the limit shall be placed on the waiting list and marked in red.
	The system shall automatically generate a unique exam ID for each exam created. This exam ID will serve as a distinct identifier for every examination, ensuring proper tracking, management, and referencing of exam-related data across the system. It will link all associated applicant registrations, documents, and results to the specific exam, allowing DCDD to manage multiple exams efficiently without confusion or overlap.
2	The system shall enable DCDD to review and verify all submitted documents and information provided by the test takers. If documents or information are incomplete or require correction, DCDD shall be able to request resubmission from the test taker. The system shall notify the test taker about the resubmission request.
3	After successful verification, the system shall mark the application as verified and automatically generate a unique registration number for the test taker. The DCDD shall be able to send notification to all the Test Takers anytime after the verification with exam time and venue via sms and email.

5.3 ABSENTEES

5.3.1 PROCESS FLOW

When the examination is conducted, test takers are expected to appear and complete all four skills: Writing, Reading, Listening, and Speaking. However, if a test taker fails to attend the examination venue or does not complete any one or more of these skills, they are considered absent. The DCDD monitors attendance during the examination period and identifies test takers who did not appear or partially appeared for the test.

Once non-attendance is confirmed, the DCDD marks the test taker as absent in the system. This action triggers an automatic update of the test taker's registration status from "Approved" or "Verified" to "Absent," clearly indicating their non-participation. The system immediately sends a notification to the test taker via SMS and email, informing them of their absentee status. This marking prevents any further processing of that registration, including band score entry by the Exam Committee and certificate generation.

The system maintains a comprehensive record of all absent test takers for administrative purposes, capturing details such as registration number, CID, scheduled exam date, and the specific skills for which they were absent. This data enables DCDD to track attendance patterns and generate reports for analysis and decision-making. Test takers marked as absent who wish to take the examination in the future must submit a fresh registration application during the next available registration window, as per DCDD's established policies.

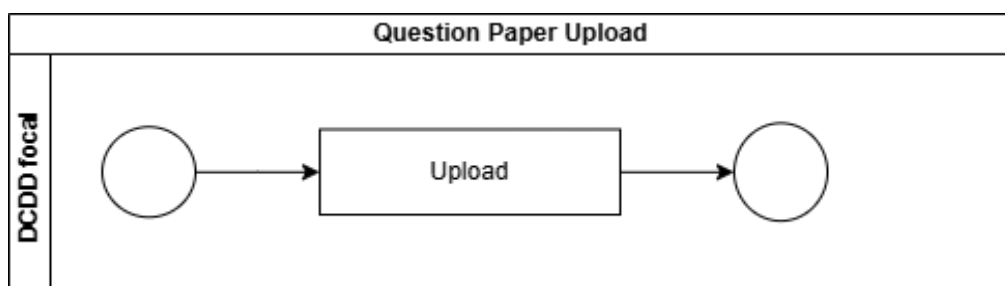
5.3.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
BR-2: UPDATE ABSENTEES (DCDD)	
1	The system shall provide DCDD with the functionality to manage and mark test takers as absent. This functionality shall only be available for test takers whose applications have been verified and approved by DCDD.

2	If a test taker fails to appear for any one or more of the four skills (Writing, Listening, Speaking, or Reading), the DCDD shall mark the test taker as "Absent" for the entire examination. Partial attendance for any subset of skills shall not be considered as completion, and the test taker shall be marked absent.
3	Once a test taker is marked as absent, the system shall automatically prevent any band score entry for that test taker. The Exam Committee shall not be able to enter, view, or submit band scores for test takers with an "Absent" status.
4	The system shall automatically update the registration status of absent test takers from "Approved" or "Verified" to "Absent" to accurately reflect their non-participation in the examination. This status change shall be recorded with a timestamp for audit purposes.
5	Upon marking a test taker as absent, the system shall automatically trigger notifications via SMS and email to inform the test taker of their absentee status.
6	The system shall maintain a comprehensive and permanent record of all absent test takers, including their registration number, CID, scheduled exam date, and the specific skill(s) for which they were absent. This information shall be accessible to DCDD for reporting, analysis, and administrative purposes.
7	Absent test takers shall not be eligible for certificate generation. The system shall automatically restrict and prevent certificate creation for any registration marked as "Absent," ensuring that only test takers who completed the examination receive certificates.
8	If a test taker marked as absent wishes to take the examination in the future, they shall be required to submit a fresh registration application during the next available registration window, subject to DCDD's policies and procedures. The system shall not carry forward or reinstate previous registrations marked as absent.

5.4 QUESTION PAPER UPLOAD

5.4.1 PROCESS FLOW



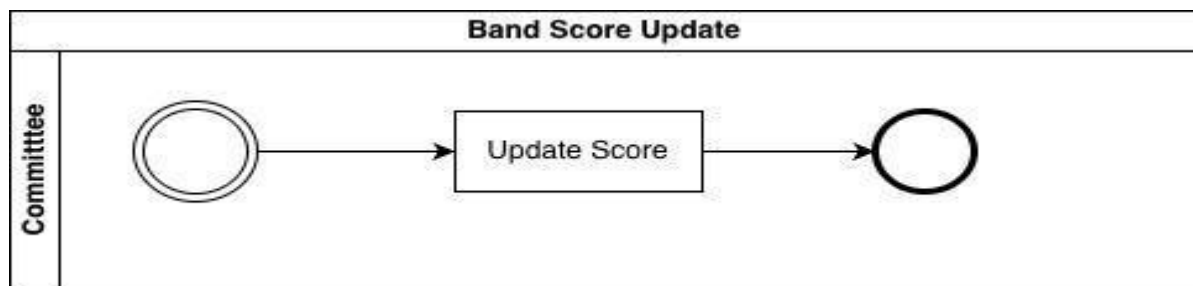
The question papers will be prepared offline. Once prepared and verified by the Exam Head, they shall be uploaded to the system along with the corresponding answer sheets.

5.4.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
QUESTION PAPER UPLOAD (EXAM HEAD)	
1	Only the DCDD Chief of Examination shall have provision to upload the question paper and the answer sheet as separate documents.
2	The uploaded question papers and answer sheets shall be encrypted to ensure confidentiality.
3	The Chief of Examination shall have the provision to access, view, and download the question papers during the scheduled exam time.
4	After the results are declared, the uploaded question papers and answer sheets shall be made available under the 'Sample Question Paper' section of the system. These documents shall then be visible and accessible to all system users for reference purposes.

5.5 BAND SCORE UPDATE

5.5.1 PROCESS FLOW



The examination papers are first evaluated manually by the designated evaluators. Following the evaluation, an evaluation committee is formed in the system, and a Committee Head is appointed. The Committee Head (Chief of Examiner) enters the band scores for each test taker into the system based on the approved marking scheme. The system records and stores the scores skill-wise for Writing, Listening, Speaking, and Reading.

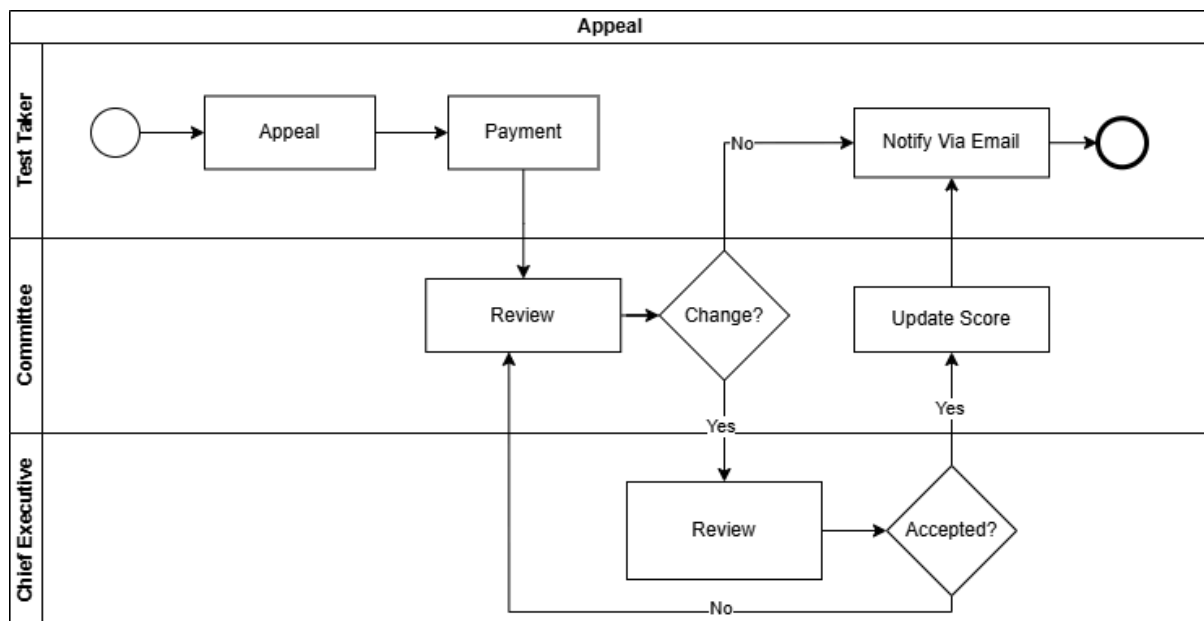
5.5.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
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BAND SCORE UPDATE (COMMITTEE)	
1	The system shall provide functionality to constitute an Exam Committee by allowing authorized users to add committee members and designate one member as the Committee Head. Only the designated Committee Head shall be granted the authority to enter and submit band scores in the system.
2	Band score entry shall be performed skill-wise for each test taker in accordance with the approved scoring format, covering Writing, Listening, Speaking, and Reading. Once the band scores are submitted, the committee members shall only have view-only access to the submitted scores, and no further modifications shall be permitted unless a formal revision process is initiated.
3	The system shall display the names of all committee members below the updated score sheet for each test taker to ensure transparency and accountability.

5.6 RE-EVALUATION

5.6.1 PROCESS FLOW



After the official declaration of examination results, a test taker who is not satisfied with their score may apply for re-evaluation through the system. The test taker can choose one or more skills for which re-evaluation is requested and must make the prescribed payment as defined in the system master data.

Once the payment is successfully completed, the appeal request is forwarded to the Examination Committee and they are notified by the system. The committee then verifies the submission and conducts the re-evaluation of the selected skill(s) offline.

If, after re-evaluation, the committee finds that a score revision is required, they submit a formal request through the system to the Chief of Examiner for approval. Until approval is granted, the system shall not allow any modification of the test taker's band score.

Upon approval by the Chief of Examiner, the system updates only the approved skill score(s). After the process is completed, the system automatically notifies the test taker of the final outcome, whether the score has been revised or remains unchanged.

5.6.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
APPEAL (TEST TAKER)	
1	Test takers shall be allowed to submit an appeal after the official declaration of results. The system shall allow appeals to be submitted against individual skills.
2	Upon submission of an appeal, the test taker shall be required to make a payment. The payment amount shall be configurable and maintained in the system master data.
3	Once payment is successfully completed, the system shall forward the appeal to the Examination Committee and notify the relevant committee members.

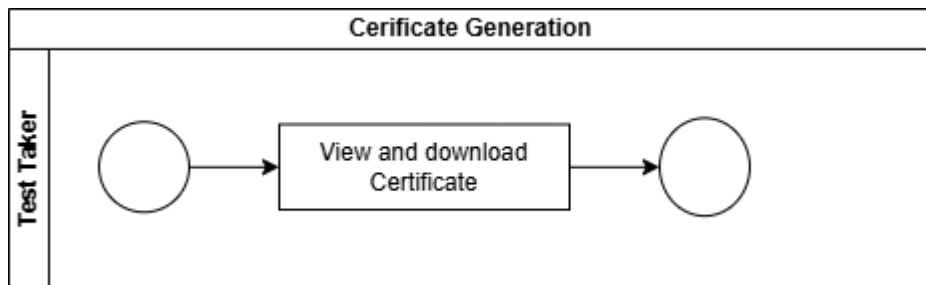
BR NO	BUSINESS LOGIC
BAND SCORE UPDATE (COMMITTEE)	
1	After reviewing the re-evaluation results, if any band score revision is required, the committee shall submit a formal request through the system to the Chief of Examiner for approval. The system shall restrict any score modification until such approval is granted.
2	The system shall allow the committee to select the specific skill or skills for which score revision is required. Only the selected and approved skills shall be updated after approval.

BR NO	BUSINESS LOGIC
VERIFICATIONS & APPROVALS (CHIEF EXECUTIVE)	

1	The system shall allow the Chief Executive to approve or reject the committee's request for score revision.
2	Only after approval shall the system permit the committee to update the band score for the selected skill(s).
3	Upon completion of the appeal and revision process, the system shall automatically notify the test taker of the final outcome, regardless of whether the band score was changed or remained unchanged.

5.7 CERTIFICATE UPDATE AND GENERATION

5.7.1 PROCESS FLOW



After the committee completes updating the band scores for each test taker, the system generates the corresponding certificates using the standard format. Each certificate includes the test taker's scores along with relevant details such as the certificate title, logos, authorized signatures, and a unique QR code. The QR code allows test takers to quickly and securely access their certificates. Test takers can also access their certificates by providing their registration number or CID, enabling them to view or download the certificate for personal use. The system records the certificate's issuance date and displays its validity period. In cases where a test taker takes another exam, the system generates a new certificate while keeping all previously issued certificates intact, accessible, and verifiable through their respective QR codes.

5.7.2 BUSINESS LOGIC

BR NO	BUSINESS LOGIC
CERTIFICATE UPDATE AND GENERATION (TEST TAKER)	

1	The system shall automatically calculate the average score for each skill (Writing, Listening, Speaking, Reading) and update the certificate accordingly once the band scores are entered by the committee.
2	All certificate fields, including Paper Size, Paper Orientation, Paper Border Image, Left & Right Logo, Certificate Title, Declaration Statement, Authorized Signature Name, Authorized Signature Seal, and Authorized Signature, shall be stored and retrieved from a master table. This ensures that any future updates to the certificate format can be applied efficiently without affecting existing certificates.
3	The system shall generate a unique QR code for each certificate. The QR code allows the test taker to securely and quickly view or verify their certificate.
4	Test takers shall be able to view and download their individual certificates using either their registration number or CID. The system shall ensure secure access so that certificates are visible only to the respective test taker.
5	The system shall allow the validity period of each certificate to be configurable and calculated from the date the band score is updated by the committee. The system shall automatically calculate and display the validity date on the certificate.
6	If a test taker takes another test, the system shall retain previous certificates and shall not replace or overwrite them. Each certificate shall remain a distinct record in the system.

6. DASHBOARD REQUIREMENTS AND REPORTS

The DSTS system shall provide comprehensive dashboards and reporting capabilities for all stakeholders, aligned with their roles, access permissions, and functional responsibilities across the examination ecosystem. Each stakeholder shall be able to view real-time summaries, key performance indicators, pending tasks, application statuses, registration notifications, score entry progress, appeal statuses, certificate generation metrics, and other relevant operational data. Dashboards shall dynamically present data generated from all modules-Registration Management, Question Paper Upload, Band Score Entry, Certificate Generation, and any newly added functions-ensuring that users always see the most up-to-date information based on their privileges.

The system shall enable users to generate, view, filter, and download reports according to their operational or analytical needs. Reports shall be accessible only to authorized users based on their existing role-based access rights. The reporting engine shall allow users to

query both standard predefined reports (such as registration summary, application verification status, band score distribution, appeal tracking, certificate validity status, examination schedules, committee performance metrics, and system audit logs) and create ad-hoc reports using any available data fields. The interface for generating custom reports shall be fully configurable, enabling selection of data sources, filters, grouping, date ranges, skill types, examination sessions, and output structure. The system shall also support exporting reports in multiple formats, including PDF, Excel, and CSV.

All dashboards and reporting elements shall update automatically when new data fields or new modules are added to the system. Newly introduced metrics, workflow fields, document statuses, score entry indicators, approval decision data, and certificate validity parameters shall appear within dashboards relevant to each stakeholder's role without requiring system redevelopment. DCDD administrators shall be able to configure which metrics are displayed for which stakeholder type, ensuring that the dashboard experience remains flexible and aligned with evolving examination policies or operational requirements.

Overall, the dashboard and reporting functionality shall serve as a centralized monitoring and analytical layer that provides transparency, supports decision-making, enhances accountability, and enables DCDD, Exam Head, Committee Members, Chief of Examiner, and test takers to track workflow progress, validate band scores, monitor certificate generation, assess appeals, and evaluate overall examination system performance efficiently.

6.1 User Authentication and Role Based Access Control

The module verifies a user's identity, ensuring that only authorized individuals can access the system. It also manages user permissions based on their assigned roles, ensuring individuals only access data and functionalities relevant to their responsibilities.

The following are the functional requirements:

1. Secure Login: The system must allow users to log in securely using the given credentials logged in the system or National Digital Identity (NDI)
2. User Management: The system must allow administrators to create, modify or delete users.
3. Role Management: The system must allow administrators to create, modify or delete roles.
4. Permission Management: The system must assign or modify permissions for each role (e.g. create, read, update, delete) for all the modules created in the system.

Additionally, the system must allow roles to be associated with one or more permissions.

5. Role Assignment: The system must allow the system administrator to assign users a role.

7. PROPOSED SOLUTIONS

7.1 SOLUTION OVERVIEW

The proposed solution introduces a comprehensive DZONGJUK (DSTS) for managing Dzongkha language proficiency examinations. The system will be web-based, mobile-responsive, and accessible through standard browsers without the need for native applications, enabling seamless access for DCDD administrators, examination officials, committee members, and test takers.

The DSTS platform will streamline the entire examination lifecycle—from registration and application verification to question paper management, band score entry, appeals processing, and certificate generation. The system will provide configurable registration windows, capacity management, automated workflows, and role-based access control to ensure efficient administration and transparent operations.

Authentication and user identity verification will align with the National Digital Identity (NDI) framework, ensuring secure access for all stakeholders. The system will automatically fetch and verify applicant information from external systems including NDI and census data to validate authenticity and reduce manual verification efforts. Test takers will be able to register, track application status, view examination details, access results, and download certificates through a user-friendly interface, while DCDD and examination officials will have comprehensive administrative controls and real-time monitoring capabilities.

The solution will incorporate master data management for configurable parameters such as registration timelines, capacity limits, certificate validity periods, certificate templates, scoring formats, and system notifications. This master-driven approach ensures flexibility to adapt to changing examination policies without requiring system redevelopment.

7.2 INTEGRATION APPROACH

The DZONGJUK system will support secure, two-way data exchange (fetch and post) with external systems such as NDI (National Digital Identity for applicant verification), census database (demographic validation), SMS gateway (notification delivery), and email services (communication management). The system will automatically retrieve and validate applicant information during registration, ensuring data accuracy and reducing manual data entry errors.

All integrations will follow secure REST API standards, with comprehensive logging for auditability and traceability. Data retrieved from external systems, such as identity verification status, demographic information, and notification delivery confirmations, will automatically update the relevant registration and examination workflows. Where APIs

are unavailable initially, the system will support interim manual verification processes, with full API automation implemented once integration endpoints are established.

The system will implement robust error handling and retry mechanisms for external service calls, ensuring that temporary connectivity issues do not disrupt critical workflows. Integration logs will be maintained for all API transactions, enabling administrators to monitor integration health, diagnose issues, and ensure data consistency across systems.

Notification services will be integrated to automatically send SMS and email alerts to test takers at key stages-application acknowledgement, verification status updates, examination schedules, result declarations, and appeal outcomes. These notifications will be triggered by system events and workflow state changes, ensuring timely communication without manual intervention.

7.3 TECHNOLOGY CONSIDERATIONS

The DSTS system will be built on modern, proven technology standards to ensure reliability, scalability, security, and maintainability:

Modular, Scalable, and Secure Web Technologies

- Microservices or modular architecture for independent scaling and deployment
- Cloud-ready, responsive web design supporting desktop, tablet, and mobile devices

High Concurrent Usage and Fast Response Times

- Optimized for high concurrent loads during registration windows and result declarations
- Database indexing, caching, and query optimization for fast performance under standard connectivity
- Asynchronous processing for bulk operations and load balancing for peak traffic management

HTTPS/TLS-Secured Data Exchanges with Role-Based Access Controls (RBAC)

- HTTPS/TLS encryption for all data transmission and encryption at rest for sensitive information
- Comprehensive role-based access control with multi-factor authentication for administrative functions
- Regular security assessments and vulnerability scanning

Comprehensive Audit Logging for All Activities and API Interactions

- Complete audit trail for all user actions and system activities with timestamps and user identity
- Detailed logging of external API interactions (NDI, census, notifications)
- Searchable, immutable logs for compliance verification and incident investigation

Support for Configurable Workflows, Master Data Management, and Dynamic Dashboards

- Master data-driven configuration for registration windows, capacity, certificates, scoring, and validity periods
- Configurable workflows with automated notifications and approval chains

Future-Ready Architecture with Minimal Redevelopment

- Extensible data model and APIs accommodating new examination types and requirements
- Plugin-based architecture for seamless feature additions
- Flexible reporting automatically incorporating new fields and metrics

8. NON-FUNCTIONAL REQUIREMENTS

CATEGORY	REQUIREMENTS
Performance	Must support at least 100 concurrent users. Average page load time should be under 3 seconds on 3G/4G networks to accommodate rural connectivity.
Scalability	The system must scale to cover all gewogs and dzongkhags without redesign. The architecture should allow seamless addition of new modules .
Availability	System uptime must be at least 99%, excluding planned maintenance. Core functions such as farmer registrations, market information must be available 24/7.
Reliability	All data and updates must be auto-saved and backed up to prevent data loss during connectivity interruptions or failures.
Security	Role-based access control (RBAC) must be enforced for all the uses. Data transmissions must use HTTPS encryption. The system must comply with data protection laws and maintain audit trails for sensitive actions.
Maintainability	The codebase must be modular to allow updates and new feature

	additions without full system redeployment. Administrative interfaces should allow configuration changes (e.g., work flows, market infrastructure ,master tables) without code changes.
Usability	The user interface must be simple for low-literacy users. Mobile-first design for Android smartphones and tablets. Minimal training should be required for all user groups.
Interoperability	The system must integrate with external APIs from DCRC, NDI, etc. It should expose APIs for future integrations with other government systems and mobile apps.
Portability	Support access via desktop browsers and Android mobile devices.
Backup & Recovery	Daily automatic backups of all data must be performed. The system should have a Recovery Time Objective (RTO) of under 4 hours for service restoration after outages.
Compliance	The system must comply with Act/Rules/Guidelines. It must align with National Digital Identity (NDI) regulations for authentication and data security.

9. RISK & MITIGATION

RISK ID	RISK DESCRIPTION	IMPACT	PROBABILITY	MITIGATION PLAN
R1	Scope Creep or Unclear/Changing Requirements: Evolving workflows, new functional demands, or unclear stakeholder expectations causing project delays, budget overruns, and delivery timeline extensions.	High	High	Adopt Agile methodology with iterative sprints and regular stakeholder reviews; maintain detailed requirements traceability matrix; implement strict change control process with impact assessments and formal approvals; freeze requirements after sign-off with defined change request procedures.
R2	Technical Integration Failures: Challenges integrating with NDI, Census Database API, or other external systems leading to authentication failures, data sync issues, or system instability.	High	Medium	Conduct thorough API documentation review and early integration testing; establish sandbox environments for testing; implement fallback mechanisms for critical integrations; maintain regular communication with NDI and DCRC technical teams; develop

				comprehensive error handling and logging.
R3	Low User Adoption and Resistance to Change: DCDD administrators, exam heads, committee members, and test takers accustomed to manual and email-based processes may resist transitioning to the digital DSTS platform, leading to continued reliance on paper records and ad-hoc coordination.	High	High	Conduct role-specific training for DCDD staff, exam committees, and test takers with hands-on walkthroughs of registration, band score entry, and certificate workflows; run a pilot examination cycle before full rollout; assign system champions within DCDD; provide helpdesk support during initial cycles; clearly communicate efficiency gains such as automated notifications, real-time status tracking, and digital certificate access.
R4	Data Integrity and Security Risks: Unauthorized access to sensitive examination data, including question papers, band scores, candidate identity records, and issued certificates, may compromise the confidentiality and integrity of the DSTS and undermine public trust in the system.	High	Medium	Enforce role-based access control (RBAC) ensuring DCDD staff, exam heads, evaluators, and test takers access only their permitted modules; encrypt all data in transit and at rest; secure question paper uploads with restricted access until examination day; maintain comprehensive audit logs of all actions; conduct regular security reviews and vulnerability assessments; comply with Bhutanese government ICT security standards.
R5	System Performance and Scalability Issues: Poor system performance during peak usage periods or inability to scale as test takers registrations and data volume grow, leading to slow response times, timeouts, and user frustration.	Medium	Medium	Design scalable cloud-based architecture with auto-scaling capabilities; conduct load testing and performance optimization; implement caching mechanisms; optimize database queries and indexing; establish performance monitoring and alerting; plan capacity based on projected growth; implement CDN for faster content delivery.
R6	Inadequate Internet Connectivity: Limited or unreliable internet access	Medium	Medium	Design lightweight interface optimized for low bandwidth; establish backup options (mobile

	hindering system usage and real-time data synchronization.			hotspots, alternative ISPs); enable basic offline data entry with later synchronization; implement data compression and local caching; train staff on connectivity troubleshooting; maintain paper backup forms for emergencies.
R7	Regulatory or Policy Changes During Implementation: Amendments to Bhutanese data protection regulations, civil service examination policies by RCSC, or electoral assessment requirements by ECB may require mid-project system modifications to DSTS workflows, scoring bands, or certificate validity rules.	Medium	Medium	Monitor regulatory and policy updates through RCSC, ECB, MoHA, and DCDD channels; design system with configurable parameters for band scores, registration windows, certificate validity, and examination rules to avoid hard-coded logic; allocate contingency budget and timeline buffer for compliance-driven changes; maintain close coordination with policy teams throughout development.
R8	Stakeholder Alignment and Coordination Challenges: Misalignment among DCDD, Divisions, and technical partners causing conflicting priorities, communication gaps, and project delays.	Medium	High	Establish project steering committee with clear governance structure; conduct regular stakeholder meetings and progress reviews; maintain centralized communication platform; define clear roles, responsibilities, and decision-making authority; document and communicate decisions promptly; implement escalation procedures for conflict resolution.

10. BRD APPROVAL & SIGN OFF

Business Requirements Document (BRD) Sign-Off

Date: [Date]

Project Name: [Project Name]

Document Overview

Provide a brief summary of the BRD, including the purpose and key objectives of the project. This section should include any critical assumptions or constraints highlighted in the document.

Attached BRD

Stakeholder Review and Approval

Please review the Business Requirements Document and provide your approval by signing below. Your signature confirms that you agree with the documented requirements and authorize the project to proceed based on these specifications.

Stakeholder	Role	Signature	Date	Comments (if any)
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]

Additional Notes

- Any additional comments or considerations can be noted here.
- Include a section for addressing any unresolved issues or action items.

Notes:

[Additional notes or comments]

Final Approval

The undersigned hereby approve the Business Requirements Document and agree to proceed with the project as specified.

Name

Role

Signature

Date

This format ensures that all relevant stakeholders have the opportunity to review and approve the document, making it clear that the project can proceed based on the agreed-upon requirements. Adjust the template to fit the specific needs and processes of your organization.